

Small Bowel Follow-Through in the Age of Cross-sectional Imaging

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Author affiliations, funding, and conflicts of interest are listed at the [end of this article](#).
See the slide presentation [here](#).

Since its inception in the early 1900s, the small bowel follow-through (SBFT) study was the only noninvasive diagnostic imaging tool for the small bowel for nearly half a century. Although CT enterography, MR enterography, and endoscopy have helped to make substantial contributions to the field of gastrointestinal medicine, SBFT remains useful in certain diagnostic scenarios. In particular, it provides real-time assessments of both function and anatomy, making it valuable in evaluating and managing adhesive small bowel obstruction and postoperative anatomy.

SBFT is performed using fluoroscopy in a stepwise fashion after the administration of oral contrast media. Barium is the preferred contrast media as it allows a more detailed evaluation. However, water-soluble contrast media should be used in place of barium when evaluating for leak or transit time in the setting of adhesive small bowel obstruction. Unlike cross-sectional imaging modalities, fluoroscopic SBFT is a dynamic examination that requires the input and active attention of a radiologist with technical expertise. The radiologist may be required to tailor the examination to answer specific clinical questions as well as make modifications in real time during the study based on any abnormalities that are observed.

Using a pattern-based approach, fluoroscopic SBFT can help diagnose a variety of conditions affecting the small bowel, including inflammatory, neoplastic, obstructive, and malabsorptive conditions. Small bowel morphologic structures can typically be characterized as one of the following diagnostic patterns: thickened folds (Fig 1), irregular or nodular folds, featureless or mucosal effacement (Fig 2), and dilated bowel with smooth regular folds. Paired with the clinical history and laboratory findings, SBFT can help to further narrow the differential diagnosis.

The water-soluble contrast challenge is a modified version of traditional fluoroscopic SBFT used to evaluate small bowel transit in special clinical situations such as small bowel obstruction due to adhesive disease or in critically ill patients. This procedure involves administering a water-soluble oral contrast agent and obtaining a series of portable abdominal radiographs to track the progress (or lack thereof) of the contrast media through the small bowel. The water-soluble contrast challenge offers high sensitivity and specificity when used as a triage tool to distinguish patients with adhesive

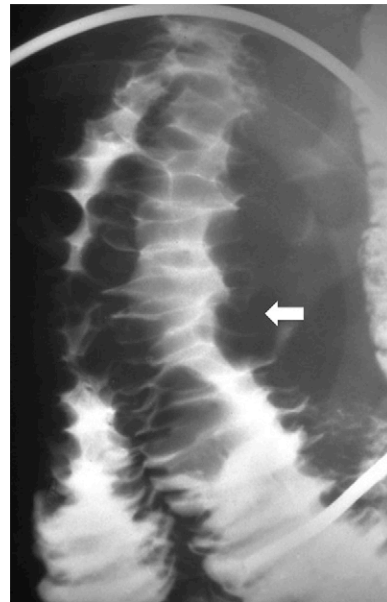


Figure 1. Intramural hematoma in a 62-year-old man taking anticoagulation medication for atrial fibrillation who presented with abdominal pain after blunt trauma. Anteroposterior fluoroscopic spot compression image from SBFT study shows smooth thickened folds (arrow) from intramural hematoma.

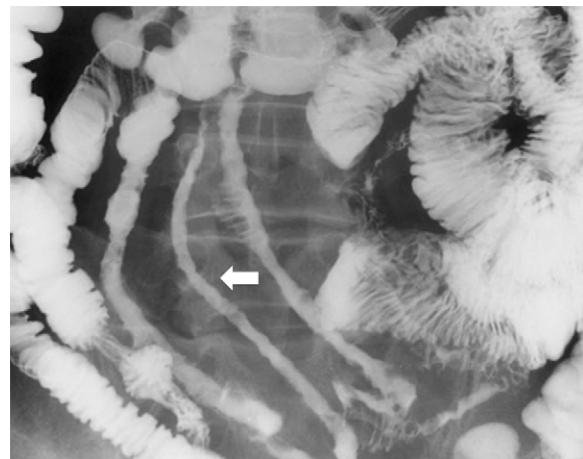


Figure 2. Graft-versus-host disease in a 45-year-old woman who underwent allogeneic hematopoietic stem cell transplant 3 years earlier for myelodysplastic syndrome and presented with chronic abdominal pain. Anteroposterior fluoroscopic image from SBFT study shows fold effacement with luminal narrowing creating a ribbonlike appearance (arrow). The findings are compatible with graft-versus-host disease.



RadioGraphics 2024; 44(7):e230135
<https://doi.org/10.1148/rg.230135>

Content Code: GI

TEACHING POINTS

- Small bowel follow-through provides both functional and anatomic assessment with low cost, wide availability, and real-time assessment tailored to the clinical question.
- Knowledge of the subtypes of abnormal bowel fold morphology and their common causes can help narrow the differential diagnosis.
- The water-soluble contrast challenge may be beneficial to guide management of patients with uncomplicated adhesive small bowel obstruction.

small bowel obstruction, which requires operative intervention, from those patients in whom nonoperative resolution of the obstruction can be expected.

Knowledge of the indications, contraindications, techniques, and pitfalls of SBFT is essential for providing appropriate patient care. The slide presentation provides an overview of the important elements of fluoroscopic SBFT and includes both radiologic and illustrative examples of normal anatomy and pathologic conditions for which the examination can provide crucial diagnostic information.

The presentation allows the viewer to navigate through a series of classic cases using a pattern-based approach, with the goal of reaffirming the value and utility of SBFT in an age of ubiquitous cross-sectional imaging.

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cepted September 12. Address correspondence to B.M. (email: mansoori@uw.edu).

Acknowledgment.—The authors would like to thank Nastaran Hosseini, MD, for their contribution to the illustrations.

Disclosures of conflicts of interest.—D.D. Compensation for speaking, including airfare and lodging, from JH Conference Planners (Meet the Masters: From Conflict to Collaboration: Facilitating Conversations with Radiologists during the Modified Barium Swallow Study; New Orleans; November 16, 2022); editorial board member of *RadioGraphics*. P.B. Editor-in-chief of *Current Problems in Diagnostic Radiology*. All other authors have disclosed no relevant relationships.

Suggested Readings

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